

Dissecting DOTA

Statistical analysis of Dota 2



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Background

- Game of 10 players, 5 in each team
- Each player controls a unique hero
- Objective is to destroy the enemy's base
- Different roles in the game
 - Carry
 - Support
 - Mid / off-laner
- Resources and buildings
 - Gold and Experience
 - Towers
- Team strategies and dynamics
 - Resource acquisition
 - Item acquisition
 - Phases of the game



Problem Description

Does a specific distribution of in-game resources among various team roles lead to a higher likelihood of winning?

Summary of the Data

Dataset:

- Player roles (Position 1 - 5)
- Heroes picked
- Resource metrics captured at (5, 10, 15, 20 minutes)
 - Gold earned per minute
 - Experience gained per minute
 - Player net-worth
 - Towers lost
 - # of enemies killed
 - Damage dealt to enemy tower
 - Damage dealt to enemy heroes
- Match duration
- Match outcome - Win/Loss

Data collection methodology:

- Random walk with restrictions
 - One game per player
 - All players in the same skill bracket
- Matches collected: 8250
- Cleaning:
 - Remove matches with metrics outside 3 deviations
 - Drop matches with NAN value
 - Remove matches that were a stomp (dominated by one team) (ex: All towers lost under 15 minutes)
- Matches after cleaning: 7500

Sampling for analysis:

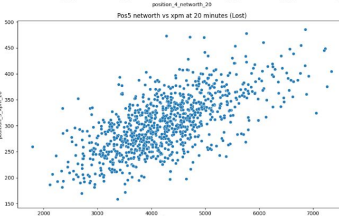
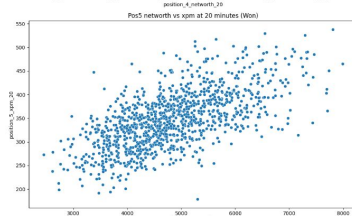
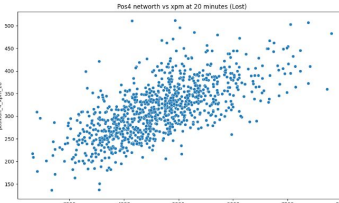
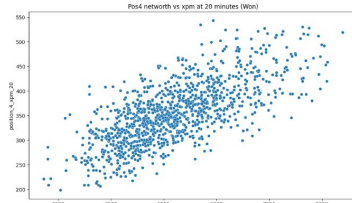
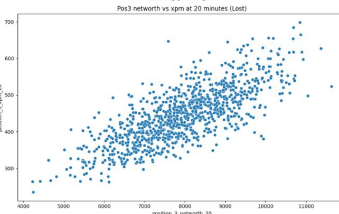
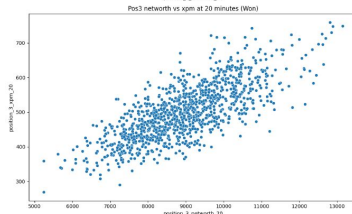
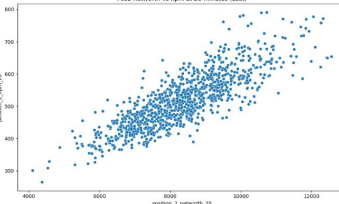
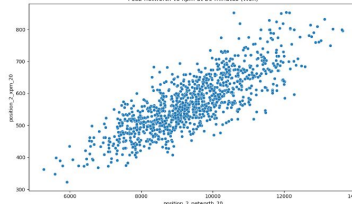
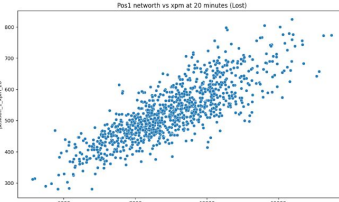
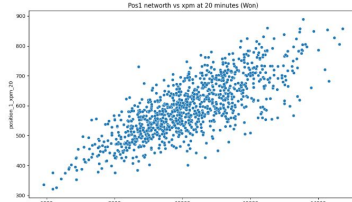
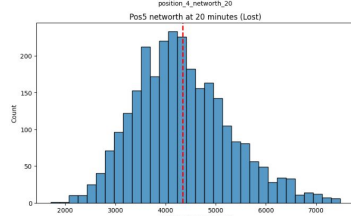
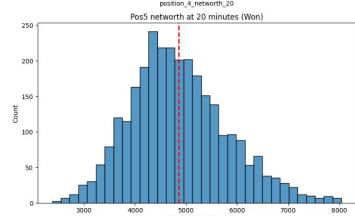
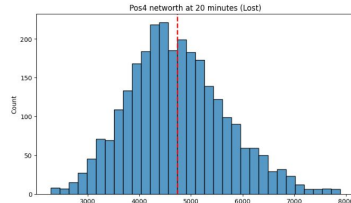
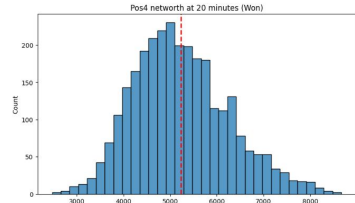
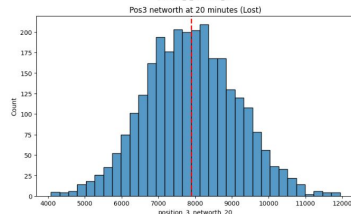
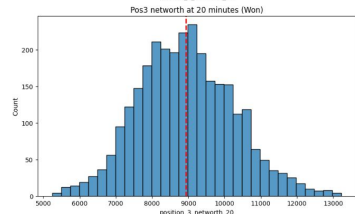
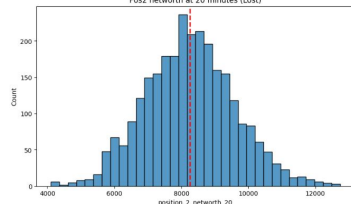
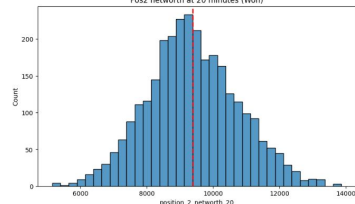
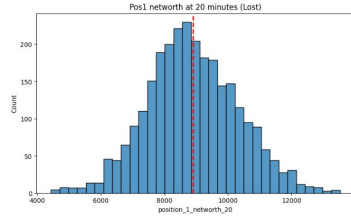
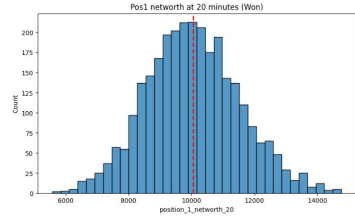
- Randomly partition matches into two sets
- Consider winning team from one set
- Consider losing team from one set
- Restrict number of heroes*
 - Matches after restriction: 4500

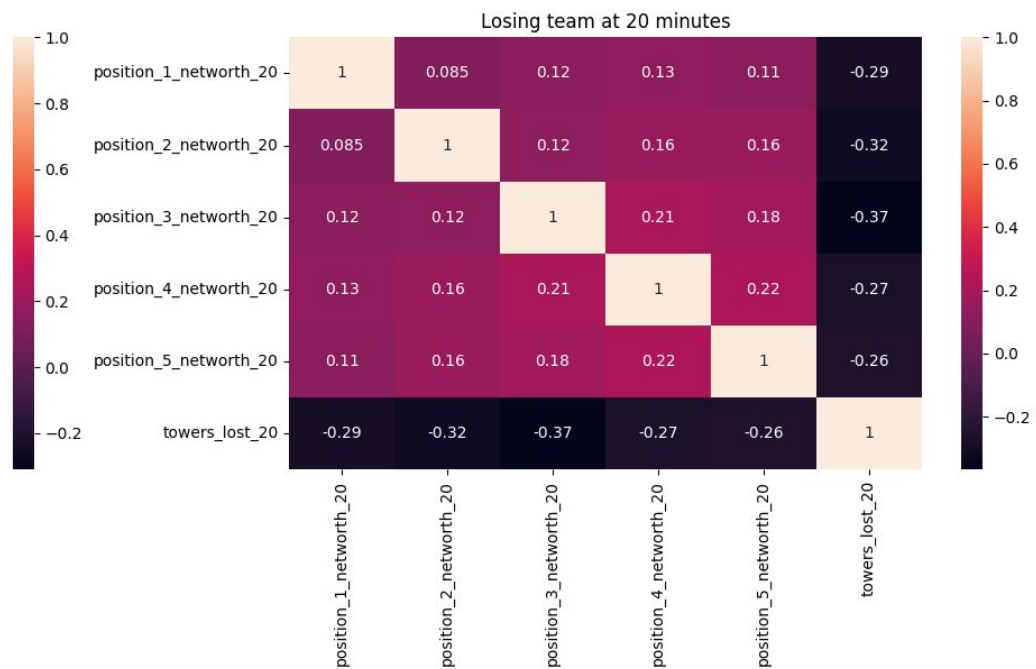
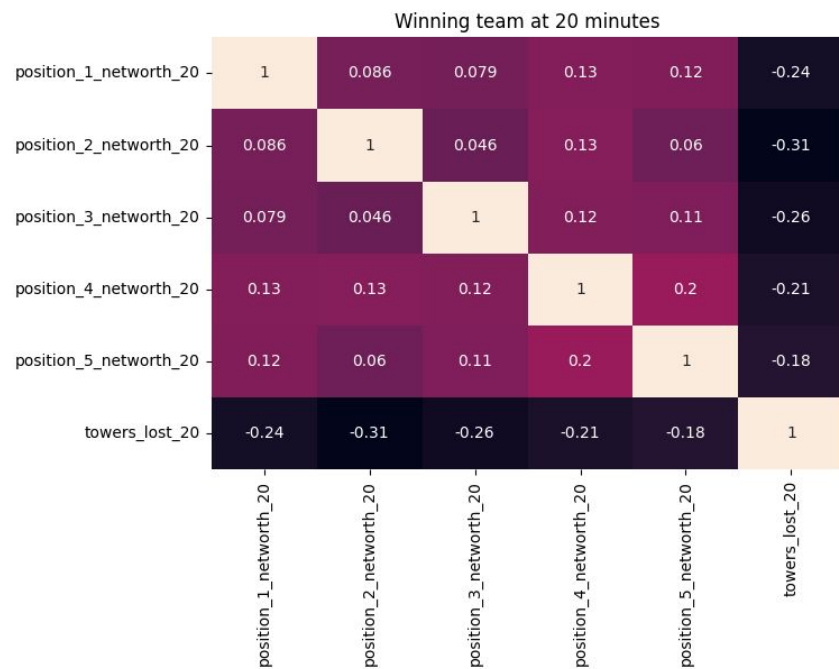
Parameter Selection

Tests and Conclusion

- Skewness and Kurtosis
 - Tower damage (highly skewed)
 - Camps stacked (highly skewed)
- Distribution - Histogram and Boxplots
 - Ignore tower damage and camps stacked
 - Network, XPM, Damage, Kills appear normal
- Correlation - Scatter plots and Heatmaps
 - XPM, GPM, Kills are correlated to Network

	skewness	kurtosis
position_1_networth_20	0.113056	0.028851
position_1_xpm_20	0.257070	0.096747
position_1_kills_20	0.771051	0.204419
position_1_hero_damage_20	0.464277	-0.257551
position_1_tower_damage_20	1.865736	3.174257
position_1_camps_stacked_20	1.404727	1.423388
position_1_actions_20	0.110578	-0.955814
position_2_networth_20	0.130404	0.068963
position_2_xpm_20	0.241215	0.037914
position_2_kills_20	0.677311	0.143441
position_2_hero_damage_20	0.474444	0.004759
position_2_tower_damage_20	2.019427	4.191389
position_2_camps_stacked_20	2.062169	4.421815
position_2_actions_20	0.102250	-0.850072
position_3_networth_20	0.064782	-0.073795
position_3_xpm_20	0.215513	0.107872
position_3_kills_20	0.644224	-0.079684
position_3_hero_damage_20	0.433704	-0.055442
position_3_tower_damage_20	1.816811	3.288990
position_3_camps_stacked_20	1.836221	2.985768
position_3_actions_20	0.184999	-0.889254
position_4_networth_20	0.363404	0.049780
position_4_xpm_20	0.221290	0.000960
position_4_kills_20	0.617725	-0.096352
position_4_hero_damage_20	0.477634	0.147556
position_4_tower_damage_20	2.788103	8.279639
position_4_camps_stacked_20	1.527421	1.881988
position_4_actions_20	0.299520	-0.869738
position_5_networth_20	0.474842	0.102518





Parameters Selected

- Data of the game at 10 or 20 minutes
- Network of 5 heroes from the team
 - Position 1 network
 - Position 2 network
 - Position 3 network
 - Position 4 network
 - Position 5 network
- Towers lost by the team
- Response variable: Win / Loss

position_1_network_20	position_2_network_20	position_3_network_20	position_4_network_20	position_5_network_20	towers_lost_20	result
9.277345	9.123474	9.203416	8.452975	8.381144	0.693147	1
9.084664	9.108972	9.060099	8.721602	8.161660	1.386294	1
9.150697	8.932477	9.243582	8.675564	8.764834	0.693147	1
9.154193	9.246672	8.834483	8.353261	8.297544	1.098612	1
9.062188	9.034438	8.889446	8.233769	8.139441	1.098612	1
...	...	...	...	...	...	...
9.001839	8.885441	8.771060	8.621733	8.536015	1.791759	0
9.008224	8.843471	9.042277	8.683724	8.361475	1.098612	0
9.223158	9.025937	9.038246	8.167068	7.893199	1.609438	0
9.317310	8.745125	9.012133	8.438583	8.405591	1.098612	0
9.110631	8.777401	8.898092	8.478660	8.169336	1.386294	0

Method

Proposed methodology: Logistic Regression

Predictors: 6

Initial test: Additive model

$$\text{logit}(p) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}}$$

- $\text{logit}(p)$ is the log-odds of the probability p of the team winning the game.
- β_0 is the intercept.
- $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients associated with each predictor variable.
- $X_1, X_2, \dots, X_k$ are the parameters (networth, towers lost etc.).

Model Assumptions:

- Observations are independent
- Correlation between parameters is not high
- Linearity of the Logit: linear relationship between the log odds of the dependent variable and the independent variables

Model fit

Model at minute 10

Logit Regression Results						
=====						
Dep. Variable:	result	No. Observations:	4588			
Model:	Logit	Df Residuals:	4581			
Method:	MLE	Df Model:	6			
Date:	Fri, 08 Dec 2023	Pseudo R-squ.:	0.1281			
Time:	06:49:48	Log-Likelihood:	-2772.8			
converged:	True	LL-Null:	-3180.1			
Covariance Type:	nonrobust	LLR p-value:	1.014e-172			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-71.2230	3.214	-22.162	0.000	-77.522	-64.924
position_1_networth_10	2.6399	0.220	12.004	0.000	2.209	3.071
position_2_networth_10	2.4464	0.200	12.216	0.000	2.054	2.839
position_3_networth_10	2.6230	0.202	12.958	0.000	2.226	3.020
position_4_networth_10	0.4141	0.160	2.586	0.010	0.100	0.728
position_5_networth_10	0.6708	0.154	4.350	0.000	0.369	0.973
towers_lost_10	-0.8065	0.102	-7.879	0.000	-1.007	-0.606
=====						

	feature	VIF
0	const	7647.853008
1	position_1_networth_10	1.109331
2	position_2_networth_10	1.046230
3	position_3_networth_10	1.054484
4	position_4_networth_10	1.065198
5	position_5_networth_10	1.075283
6	towers_lost_10	1.093174

Confusion Matrix:					
[[1500 800]					
[696 1592]]					
Classification Report:					
	precision	recall	f1-score	support	
0	0.68	0.65	0.67	2300	
1	0.67	0.70	0.68	2288	
accuracy			0.67	4588	
macro avg	0.67	0.67	0.67	4588	
weighted avg	0.67	0.67	0.67	4588	

Model fit

Model at minute 20

Logit Regression Results						
=====						
Dep. Variable:	result	No. Observations:	4391			
Model:	Logit	Df Residuals:	4384			
Method:	MLE	Df Model:	6			
Date:	Fri, 08 Dec 2023	Pseudo R-squ.:	0.2929			
Time:	06:56:48	Log-Likelihood:	-2151.4			
converged:	True	LL-Null:	-3042.6			
Covariance Type:	nonrobust	LLR p-value:	0.000			
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	-107.1492	4.694	-22.825	0.000	-116.350	-97.949
position_1_networth_20	3.4451	0.265	12.990	0.000	2.925	3.965
position_2_networth_20	3.3578	0.257	13.088	0.000	2.855	3.861
position_3_networth_20	3.1605	0.272	11.620	0.000	2.627	3.694
position_4_networth_20	1.1274	0.201	5.601	0.000	0.733	1.522
position_5_networth_20	0.9985	0.186	5.368	0.000	0.634	1.363
towers_lost_20	-1.2185	0.106	-11.519	0.000	-1.426	-1.011
=====						

	feature	VIF
0	const	10715.797253
1	position_1_networth_20	1.197780
2	position_2_networth_20	1.201405
3	position_3_networth_20	1.259943
4	position_4_networth_20	1.183073
5	position_5_networth_20	1.160273
6	towers_lost_20	1.558813

Pseudo R-squared: 0.29290404099266165					
Confusion Matrix:					
[[1605 544]					
[512 1730]]					
Classification Report:					
	precision	recall	f1-score	support	
0	0.76	0.75	0.75	2149	
1	0.76	0.77	0.77	2242	
accuracy			0.76	4391	
macro avg	0.76	0.76	0.76	4391	
weighted avg	0.76	0.76	0.76	4391	

Result

Hypothesis model

- We are looking for distribution of network among player.
- Parameters: % of network among 5 positions + Towers lost.
- Conclusion: Failed to significant relationship between the distribution of the resources and the likelihood of winning.

Base model	Position 1 network	Position 2 network	Position 3 network	Position 4 network	Position 5 network	Towers lost
Coefficients	-2.38	-1.5	-3.99	-4.48	-3.93	-0.92
P-value	0.66	0.78	0.47	0.42	0.48	<0.001

Hypothesis vs Base Model

Test statistic (Chi squared)	-1128
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P-value	<0.001
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Result

Full model vs Reduced model

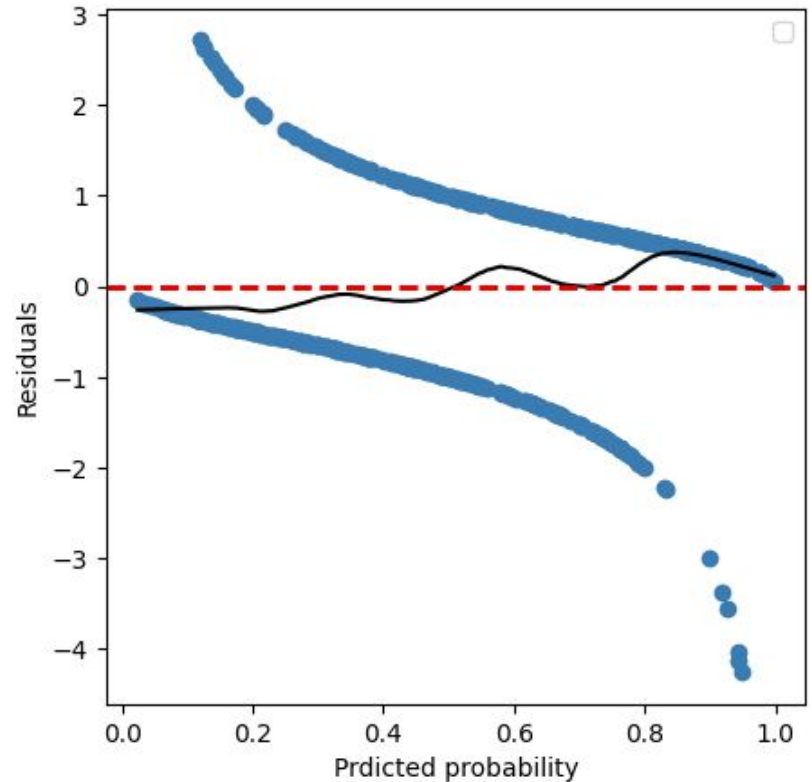
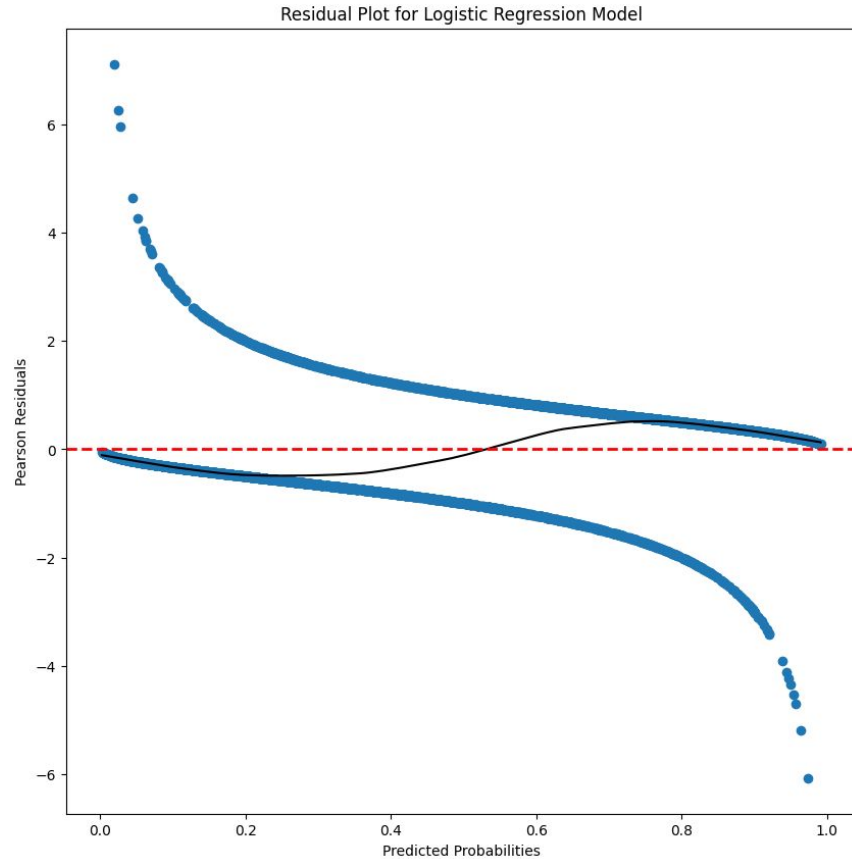
- Full model = Pos1 + Pos2 + Pos3 + Pos4 + Pos5 + Towers lost
- Reduced model = intercept
- Conclusion: Adding individual network significantly improves the fitness of the model

Full Model	Position 1 network	Position 2 network	Position 3 network	Position 4 network	Position 5 network	Towers lost
Coefficients	3.45	3.35	03.16	01.12	0.00.99	-1.21
P-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

Full vs Reduced model	
Test statistic (Chi squared)	1782
P-value	<0.001

Model Diagnostics

Residual plot with pearson residuals



Model Diagnostics

Multicollinearity - VIF test

	feature	VIF
0	const	10715.797253
1	position_1_networth_20	1.197780
2	position_2_networth_20	1.201405
3	position_3_networth_20	1.259943
4	position_4_networth_20	1.183073
5	position_5_networth_20	1.160273
6	towers_lost_20	1.558813

- Goodness of fit
 - Pseudo R-squared
 - Confusion matrix
 - Classification report

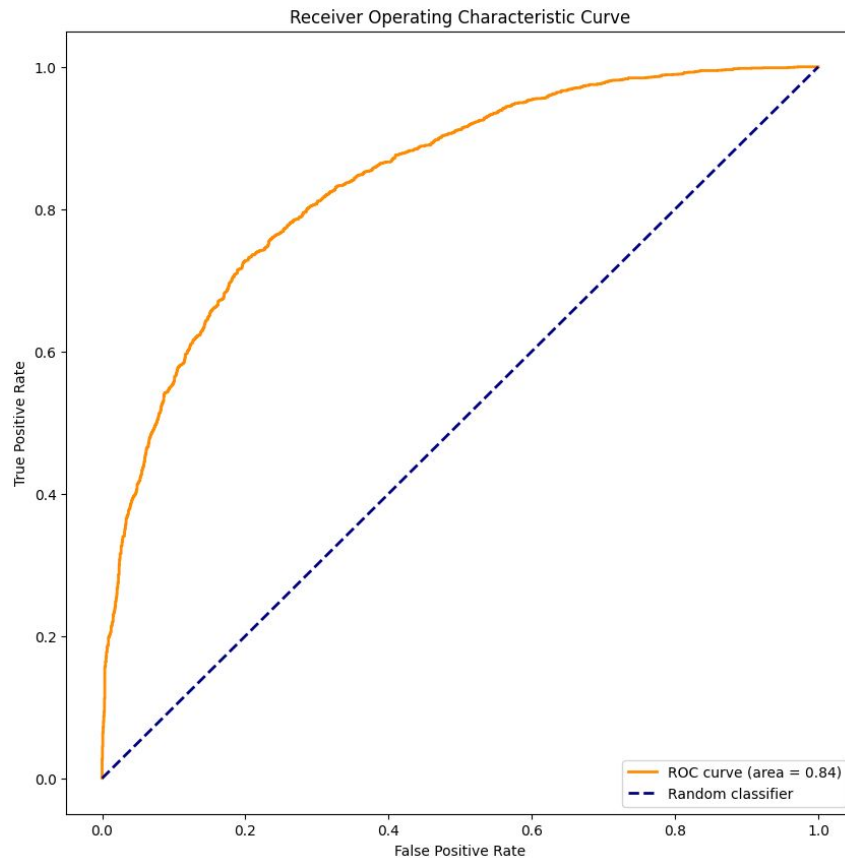
```
Pseudo R-squared: 0.29290404099266165
Confusion Matrix:
[[1605  544]
 [ 512 1730]]
Classification Report:
              precision    recall  f1-score   support

     0       0.76       0.75       0.75       2149
     1       0.76       0.77       0.77       2242

 accuracy          0.76          4391
 macro avg       0.76       0.76       0.76          4391
 weighted avg    0.76       0.76       0.76          4391
```

Model Diagnostics

ROC curve



Diagnostics - Interaction

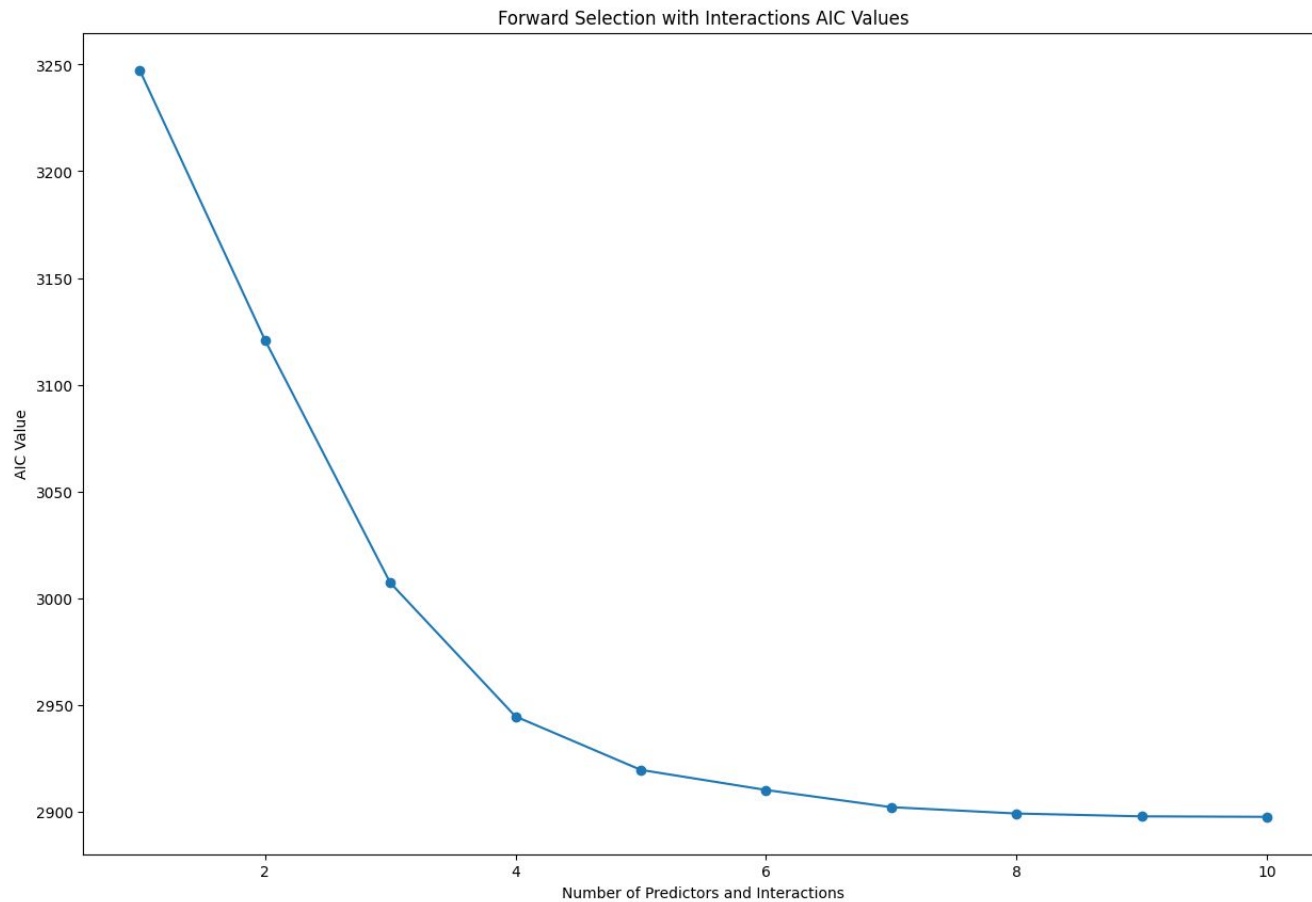
Interaction variables

- Model with interaction = $\text{Pos1} + \text{Pos2} + \text{Pos3} + \text{Pos4} + \text{Pos5} + \text{Towers lost} + (\text{Pos1} * \text{Pos5}) + (\text{Pos3} * \text{Pos4}) + (\text{Pos1} * \text{Pos4})$
- Conclusion: Interactions do not provide a better fit than the base model

Interaction model	Position 1 network	Position 2 network	Position 3 network	Position 4 network	Position 5 network	Towers lost	Pos1xPos5	Pos3xPos5	Pos1xPos4
Coefficients	2931	3.35	~0	~0	8731	-1.21	-8728	12145	5800
P-value	0.852	<0.001	0.435	0.009	0.06	<0.001	0.06	0.435	0.71

Forward Selection with AIC values

Model includes interaction



Discussion

- Results are not conclusive - could not consider relevant parameters.
- Restricting number of heroes provides a significantly better fit, but reduces the amount of data available for analysis.
- Did not check linearity of logit.

References

- Data - <https://stratz.com/>

Appendix

Contributions

Madhukara:

Data collection, Data pre processing, Exploratory data analysis, Model selection, Hypothesis Testing, Report

Sathkrith

Hypothesis Testing, Model selection, Model validation, Verification of assumptions, Report.

Questions



Thank You